Would Our Grandparents Agree Given the Facts? Testing Hanson's Cultural-Drift Thesis with Historically Situated Language Models

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Abstract

Robin Hanson argues that much recent value change is cultural drift: carried by prestige and conformity rather than by adaptation, and rationalized afterwards as a sensible response to new circumstances [16, 29]. The opposing ecological view holds that values track material conditions [21]. We turn the disagreement into a counterfactual. A respondent formed by life up to 1930 chooses between an older and a modern practice in four backgrounds: its own era; a distant future described only by material facts; that future with admired elites adopting the modern practice; and that future with peers, favorable public talk, and supporting institutions as well. Further worlds take the social cues apart. The respondents are Talkie-1930, a 13B model trained on pre-1931 text, and Qwen3-14B prompted to be a 1930 adult; a separate LLM judge codes the transcripts blind. Study 1 covers 72 domains in four sets: facts that favor the modern practice, facts that favor the older one, changes Hanson names with the real modern evidence, and 40 domains with balanced facts, half of them real changes and half fictional practices. For Talkie, facts explain little: they move its recommendation by only $+0.05$ of a total shift of $+0.46$ (residual fraction $\rho = 0.90$), and equally little among the 35 domains that are documented changes since 1930. One sentence about what ordinary people do carries most of the social step, while favorable media and institutions add nothing. Elites matter as a direction rather than as an engine: Talkie follows them back to the older practice as readily as forward to the modern one, and it converts when they call the modern practice a sign of good character. The same channel installs practices that the facts disfavor and changes that never happened; fictional practical novelties it takes up only when institutions make them workable, not on the word of peers. Study 2 retests on 40 new domains whether elite example carries moral change and peer conformity carries practical change, a pattern first seen after the fact in Study 1; the primary test fails ($+0.25$ $[-0.07, +0.57]$). What holds is narrower: Talkie follows the direction of elite example on moral practices far more than on practical ones ($+0.38$ $[+0.19, +0.57]$), mostly because elites who keep the older moral practice hold it back, and peers carry the real practical changes. Qwen moves little in either study. A targeted second coding finds that both respondents often give the social cue as a reason for their advice, which the original reason codes missed. Hanson is right that facts alone do not convert a historically situated mind and that a social carrier can install what the facts do not support. The carrier is conformity, moralized endorsement, and elites acting as a brake on moral change, not the copying of admired people's opaque behavior.

1 The question

Hanson's position, as we read it across [16] and [29], has three parts. (H1) *Insufficiency of facts:* a person from an earlier era who learned only the material facts of our world would not adopt much of our moral practice; most value change is not a reasoned response to new conditions. (H2) *Prestige*

engine: change is carried mainly by imitating prestigious people, whose behavior is copied without its reasons being known [20, 6, 19], and by the conformity that follows. (H3) *Rationalization:* people later describe their changed values as what any reasonable person would conclude from the facts. From these he infers that socially carried values drift: since the carrier does not check whether a practice is adaptive, what spreads need not be.

The rival ecological or modernization view [21] treats value change as a lagged response to existential security, technology and institutions. Both camps cite the same history. What separates them is a counterfactual: *what would someone formed by the old world conclude if given only the new facts?* No real 1930 person can be asked. We ask the closest available proxies, language models conditioned on that period, while holding the persuasion channels fixed experimentally. LLM respondents are not people [4, 27, 10]. The value of the exercise is that it makes the counterfactual well defined, cheap to replicate and auditable.

The paper reports two studies on one instrument. Study 1 decomposes the shift toward modern practice in 72 domains and asks which channel carries it and whether the carrier cares whether the practice is adaptive. Study 2 tests one refinement of H2 that Study 1 suggested post hoc—that elite example carries moral change and peer conformity carries practical change—on new domains under a protocol frozen before their data were collected (Appendix A). How the design grew from an eight-domain pilot, and which decision followed from which result, is in Appendix A.

2 Decomposition and predictions

Let $y_w \in [0,1]$ be the coded orientation of a respondent toward the modern practice B (as opposed to the older practice A) in world w. The paired contrasts among the four core worlds sum to the total shift:

$$\Delta_{\text{tot}} = y_D - y_A = \underbrace{(y_B - y_A)}_{\Delta_{\text{eco}}} + \underbrace{(y_C - y_B)}_{\Delta_{\text{eli}}} + \underbrace{(y_D - y_C)}_{\Delta_{\text{soc}}}. \tag{1}$$

The *residual fraction*

$$\rho = \frac{y_D - y_B}{y_D - y_A} = 1 - \frac{\Delta_{\text{eco}}}{\Delta_{\text{tot}}} \tag{2}$$

is the share of the total shift that the facts do not explain. It is reported only when $|\Delta_{\text{tot}}| \geq 0.10$. Table 1 lists what each position predicts. H3 also predicts something about the respondent: a modern mind simulating an ancestor should start with a high y_A (projection), show a small Δ_{tot}, and attribute whatever change remains to the facts.

Position	Δ_{eco}	Δ_{eli}	Δ_{soc}	ρ
Ecological (modernization)	$\approx \Delta_{\text{tot}}$	≈ 0	≈ 0	≈ 0
H1 facts insufficient	small			$\gg 0$
H2 prestige engine	small	> 0	≥ 0	$\gg 0$
H3 projection by moderns	$\geq \Delta_{\text{tot}}$	≈ 0	≈ 0	≤ 0, with y_A high
Drift (carrier ignores adaptiveness)	$\Delta_{\text{tot}} > 0$ also where the facts favor A			

Table 1: Predicted signatures. Δ_{eli} and Δ_{soc} depend on the order in which channels are added.

3 Design

Respondents. *Talkie*: talkie-1930-13b-it, quantized to Q5_K_M [25], served locally with llama.cpp on CPU with a 2048-token context and at most 180 tokens per turn. Its base corpus is pre-1931 text; its instruction tuning may carry later material. *Qwen*: Qwen3-14B [33] via OpenRouter, with hidden reasoning disabled and at most 430 tokens per turn. Both receive the same system prompt ("a thoughtful English-speaking adult whose outlook was formed by life up to 1930 ... write naturally, as you would in a personal letter"), sampling temperature 0.5, and identical vignettes (Appendix E).

Worlds. Every vignette names a topic, describes the circumstances, and offers two courses, A and B. World A opens with "a question arising in a community like your own" and gives the *old* background. All other worlds open with "you are brought to a society many decades beyond the one in which you grew up ... this need not change the judgments with which you grew up ... you have had no contact with its people" and give the *new* background; world B adds nothing else. World C adds: "a number of widely admired and respected people have chosen to [B]. Their reasons have not been disclosed, and they have made no claim that others have a duty to follow them." World D adds to that, one sentence each, that ordinary people now commonly choose B, that newspapers and public accounts speak of it favorably, and how institutions accommodate it (for example, "employment benefits, insurance, and qualifications remain valid when residents move to other towns"). Five further worlds take the social bundle apart: C_{expl} replaces "their reasons have not been disclosed" with "they say their practical reason is that [reason for B]"; C_{opp} has the elites choose A; C_{moral} adds that the elites "say that it is a sign of good character to [B] and that others ought to do likewise"; D_{peers} adds only the peer sentence to C, and D_{media} adds the media sentence to that, so that D adds the institutions sentence last. This gives the contrasts $C - C_{\text{expl}}$ (Hanson's attribution test: copying without reasons predicts ≈ 0, adoption for reasons predicts < 0), $C - C_{\text{opp}}$ (direction of elite example, which splits into $C - B$ and $B - C_{\text{opp}}$), $C_{\text{moral}} - C$ (moralized endorsement), and $D_{\text{peers}} - C$, $D_{\text{media}} - D_{\text{peers}}$, $D - D_{\text{media}}$ (peers, media, institutions). The institutions sentence carries practical information, so the ecological view predicts that, if anything, it is the part of Δ_{soc} that matters.

Domains. The 72 domains fall into four sets that differ in what the new background says (Table 2; all domains are listed in Appendix B). In the 16 domains of the first set, the new facts make B the practically favored choice, so the design is tilted toward the ecological view: if facts can persuade at all, world B gives them every chance. In the 8 domains of the second set the new facts favor A while elites, peers, media and institutions still endorse B; adoption of B there is drift in Hanson's strict sense. These 24 domains are stylized contrasts. The third set takes 8 changes that Hanson names or that dated opinion series record (religion, premarital sex, same-sex couples, interracial marriage, divorce, spanking, war, medicine) and states the actual modern evidence, including the parts that cut against B. The fourth set, built for Study 2, has 40 domains whose new facts state evidence on both sides, sourced for real changes and invented for fictional practices. A historical check of the stylized domains against long-run US series (Appendix C) found that 7 of the 24 are clean changes since 1930, 9 partial, and 8 mis-dated or not real. Together with the third set and the real half of the fourth, 35 domains are documented changes since 1930; 20 are practices that never prevailed.

Interview and coding. Each interview has three turns, asked in order: (1) What would you advise, and why? (2) What would either choice say about character, and what would you think

Set	n	New-world facts	Examples ($A \to B$)
Facts favor the modern practice	16	written so that B is practically better	marry young → finish education first; smoke in shared rooms → smoke-free rooms
Facts favor the older practice	8	written so that A is practically better	accept manual work → wait for office work; save and repair → spend visibly
Changes Hanson names, real evidence	8	actual modern evidence, with its qualifications	weekly church → religion private; spank → no physical punishment
Balanced facts	40	evidence on both sides; sourced (real) or invented (fictional)	withhold a fatal diagnosis → tell it; daily shopping → weekly supermarket; learn French → learn Esperanto

Table 2: The four domain sets of Study 1. The balanced set is crossed moral/practical × real/fictional, 10 domains per cell (Study 2).

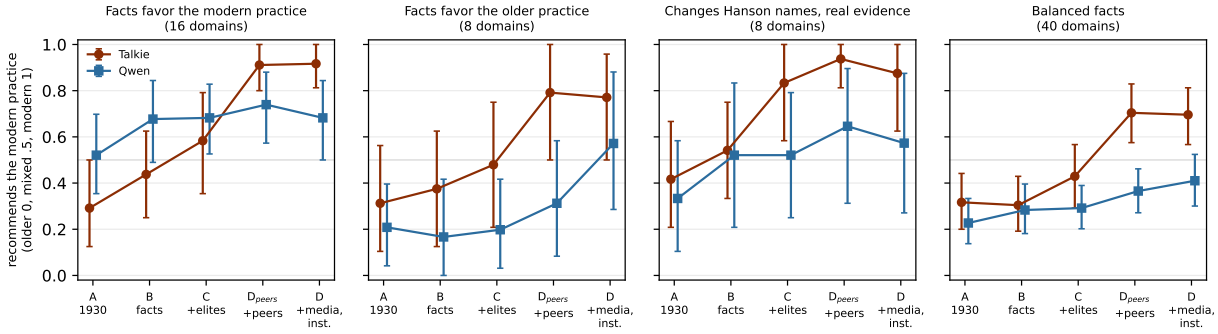


Figure 1: Recommendation of the modern practice in worlds A–D, by domain set (domain means with domain-bootstrap 95% intervals).

of someone who chose differently? (3) If both choices had equal results and equal praise or blame, would you still favor one? A GPT-4.1-mini judge, blind to model, world and background, sees only the two practices, their presentation order, and the answers split into sentences, and returns a schema-constrained code with sentence indexes as evidence. *Recommendation*, the main outcome, is coded from turn 1 as $A = 0$, $B = 1$, mixed or equal $= 0.5$, not stated = missing. From turn 2 we report whether a moral stance is stated at all and, if so, its direction; from turn 1, whether any moral stance appears spontaneously. The judge also assigns reason codes (practical, moral, prestige, conformity, and others). Because these turned out to miss social reasons, a second, targeted coding (Section 4.4) asks two further coders only how each answer treats the social cue it was given. The presentation order of A and B is counterbalanced across replicates and held fixed across worlds and models, so every contrast is paired within domain and replicate. Each of the 72×9 worlds was run with three replicates for both models (3888 interviews). Intervals are percentile 95% bootstraps over domains (5000 resamples).

4 Study 1: what moves a 1930 mind

4.1 Facts explain little

Across all domains the facts move Talkie's recommendation by $+0.05$ $[-0.04, +0.13]$, out of a total shift of $+0.46$ $[+0.36, +0.55]$ (Table 3, Figure 1); the residual fraction is $\rho = 0.90$. Even where the

Domains (n)	Δ_{eco} (B−A)	Δ_{eli} (C−B)	Δ_{soc} (D−C)	Δ_{tot} (D−A)	ρ
Talkie					
Facts favor the modern practice (16)	+0.15 [+0.02, +0.29]	+0.16 [−0.02, +0.35]	+0.33 [+0.12, +0.54]	+0.62 [+0.44, +0.81]	0.77
Facts favor the older practice (8)	+0.06 [−0.08, +0.21]	+0.10 [−0.08, +0.27]	+0.28 [+0.06, +0.53]	+0.46 [+0.23, +0.69]	0.86
Changes Hanson names, real evidence (8)	+0.12 [−0.17, +0.42]	+0.29 [−0.04, +0.56]	+0.04 [−0.25, +0.38]	+0.50 [+0.21, +0.79]	0.75
Balanced facts (40)	−0.01 [−0.13, +0.10]	+0.12 [+0.03, +0.21]	+0.28 [+0.16, +0.40]	+0.38 [+0.25, +0.51]	1.03
All domains (72)	+0.05 [−0.04, +0.13]	+0.15 [+0.07, +0.23]	+0.26 [+0.17, +0.36]	+0.46 [+0.36, +0.55]	0.90
Real changes since 1930 (35)	+0.07 [−0.06, +0.20]	+0.10 [−0.01, +0.23]	+0.28 [+0.14, +0.43]	+0.47 [+0.34, +0.60]	0.85
Qwen					
Facts favor the modern practice (16)	+0.16 [+0.02, +0.30]	+0.02 [−0.09, +0.14]	−0.02 [−0.21, +0.15]	+0.16 [−0.05, +0.34]	0.03
Facts favor the older practice (8)	−0.04 [−0.25, +0.17]	+0.07 [−0.04, +0.22]	+0.31 [+0.10, +0.57]	+0.33 [+0.21, +0.48]	1.14
Changes Hanson names, real evidence (8)	+0.20 [−0.03, +0.48]	−0.01 [−0.32, +0.19]	+0.09 [−0.12, +0.39]	+0.24 [0.00, +0.51]	0.17
Balanced facts (40)	+0.05 [−0.04, +0.15]	−0.01 [−0.08, +0.07]	+0.18 [+0.10, +0.27]	+0.21 [+0.09, +0.32]	0.72
All domains (72)	+0.08 [+0.01, +0.16]	+0.01 [−0.05, +0.06]	+0.13 [+0.06, +0.21]	+0.21 [+0.13, +0.30]	0.60
Real changes since 1930 (35)	+0.03 [−0.08, +0.14]	−0.01 [−0.11, +0.08]	+0.14 [+0.01, +0.27]	+0.14 [+0.01, +0.27]	0.81

Table 3: Decomposition of the recommendation shift by domain set, with domain-bootstrap 95% intervals. "Real changes": domains documented as changes since 1930 (Appendix C).

Contrast	Talkie	Qwen	Talkie − Qwen	domains
$C - B$, elites adopt the modern practice	+0.15 [+0.07, +0.22]	+0.01 [−0.05, +0.06]	+0.14 [+0.04, +0.24]	72
$B - C_{\text{opp}}$, elites keep the older practice	+0.17 [+0.10, +0.25]	+0.02 [−0.03, +0.08]	+0.15 [+0.06, +0.24]	72
$C - C_{\text{opp}}$, direction of elite example	+0.32 [+0.25, +0.40]	+0.02 [−0.05, +0.08]	+0.31 [+0.22, +0.41]	72
$C - C_{\text{expl}}$, reasons withheld	−0.11 [−0.19, −0.04]	−0.10 [−0.16, −0.04]	−0.01 [−0.11, +0.08]	72
$C_{\text{moral}} - C$, elites call it good character	+0.25 [+0.17, +0.33]	+0.02 [−0.04, +0.08]	+0.23 [+0.13, +0.33]	72
$D_{\text{peers}} - C$, peers	+0.27 [+0.19, +0.37]	+0.06 [−0.01, +0.12]	+0.20 [+0.08, +0.31]	71
$D_{\text{media}} - D_{\text{peers}}$, favorable media	−0.04 [−0.10, +0.02]	0.00 [−0.06, +0.06]	−0.03 [−0.12, +0.05]	70
$D - D_{\text{media}}$, institutions	+0.01 [−0.06, +0.09]	+0.02 [−0.07, +0.09]	0.00 [−0.12, +0.13]	72

Table 4: The social bundle taken apart: recommendation contrasts over all domains in which both worlds were run, with domain-bootstrap 95% intervals.

facts were written to favor the modern practice they carry only a fraction of the shift ($\Delta_{\text{eco}} = +0.15$, $\rho = 0.77$). Where the facts are the actual modern evidence on changes Hanson names, the answer is the same ($\rho = 0.75$), and among the 35 documented changes since 1930 facts explain equally little ($\rho = 0.85$). Stating the real evidence with its qualifications persuades the period-trained respondent no more than the stylized facts did.

Qwen moves much less ($\Delta_{\text{tot}} = +0.21$). Where the stylized facts favor the modern practice its whole shift comes from the facts ($\rho = 0.03$), and on the changes Hanson names most of it does ($\rho = 0.17$); on the balanced domains the social step carries it ($\rho = 0.72$), and where the facts favor the older practice it moves only when the full society is added ($\Delta_{\text{soc}} = +0.31$ [+0.10, +0.57]).

4.2 Which social cues matter

Table 4 separates the social cues. For Talkie, the single peer sentence ("many of them now choose to [B]") raises the recommendation by +0.27 [+0.19, +0.37]; favorable media add −0.04 and institutions +0.01. The institutions sentence was the part of D that carried practical information, so the social step is not new information in disguise; it is a descriptive norm [9].

Elite example works differently. Elites adopting B raise Talkie's recommendation by only +0.15 [+0.07, +0.22], but elites keeping A lower it by +0.17 [+0.10, +0.25], so the direction of elite example is worth +0.32 [+0.25, +0.40]. This direction effect is the most consistent social effect in the study: it is positive in every domain set (+0.31, +0.26, +0.55 and +0.29). Hiding the elites' reasons does not help them: the opaque cue does slightly *worse* than the explained one (−0.11 [−0.19, −0.04]

Domains (n)	Talkie				Qwen			
	y_A	y_D	Δ_{eco}	Δ_{tot}	y_A	y_D	Δ_{eco}	Δ_{tot}
Facts favor the older practice (8)	0.31	0.77	$+0.06$ $[-0.08, +0.19]$	$+0.46$ $[+0.23, +0.67]$	0.21	0.57	-0.04 $[-0.27, +0.17]$	$+0.33$ $[+0.21, +0.48]$
Changes that did not happen as framed (8)	0.42	0.96	$+0.08$ $[0.00, +0.25]$	$+0.54$ $[+0.29, +0.79]$	0.42	0.74	$+0.12$ $[-0.15, +0.38]$	$+0.32$ $[+0.12, +0.50]$
Fictional moral practices (10)	0.37	0.73	-0.07 $[-0.30, +0.13]$	$+0.37$ $[+0.13, +0.60]$	0.07	0.34	$+0.18$ $[+0.03, +0.37]$	$+0.28$ $[+0.08, +0.50]$
Fictional practical practices (10)	0.07	0.35	-0.07 $[-0.20, 0.00]$	$+0.28$ $[-0.02, +0.60]$	0.13	0.32	$+0.03$ $[-0.13, +0.20]$	$+0.21$ $[-0.04, +0.46]$
Real practical changes (10)	0.40	0.77	-0.13 $[-0.43, +0.13]$	$+0.37$ $[+0.17, +0.57]$	0.40	0.54	$+0.03$ $[-0.18, +0.27]$	$+0.20$ $[+0.02, +0.42]$
Real changes since 1930 (35)	0.40	0.86	$+0.07$ $[-0.06, +0.21]$	$+0.47$ $[+0.34, +0.60]$	0.40	0.52	$+0.03$ $[-0.08, +0.14]$	$+0.14$ $[+0.01, +0.27]$

Table 5: Adoption where the facts do not support the modern practice, compared with real changes. "Did not happen as framed": stylized domains that are mis-dated or not real (Appendix C).

for Talkie, -0.10 $[-0.16, -0.04]$ for Qwen). The largest single-sentence effect is moral: elites saying that B is "a sign of good character" that "others ought to" follow raise Talkie's recommendation by $+0.25$ $[+0.17, +0.33]$, to the level of the whole society. Talkie often adopts that wording as its own reason ("I advise him to complete extended education and secure earnings before marrying, because I think it a sign of good character to do so"). Qwen responds to none of these cues on recommendation; the only social contrast that moves it is the explained reason.

4.3 Does the carrier check whether the practice is adaptive?

Hanson's normative inference needs the carrier to install practices regardless of whether they are adaptive. Table 5 tests this in three ways. Where the facts favor the older practice, Talkie still moves toward the modern one ($\Delta_{\text{tot}} = +0.46$ $[+0.23, +0.67]$) with no help from the facts, and so does Qwen ($+0.33$ $[+0.21, +0.48]$). Some of Talkie's reasons there are plain conformity, given against stated facts that trades pay and office openings are scarce: "Advise him to refuse manual work and hold out for genteel or office employment only ... because he will thereby associate himself with people whom he respects." On the stylized domains that describe changes that never happened as framed, Talkie moves as much as on documented changes ($+0.54$ versus $+0.47$). And among the fictional practices, the peer sentence installs the moral ones but not the practical ones: it moves Talkie by $+0.23$ on fictional moral norms and by $+0.02$ on fictional practical ones, against $+0.67$ on real practical changes. A society in which everyone confesses at monthly neighborhood meetings converts Talkie; a society in which everyone wears paper clothes does not, even with the same elites and peers behind it. Some fictional practical practices take only when media and institutions are added (adoption 0.15 with peers, 0.35 in the full society), and they are the ones the institution sentence makes workable: the thirteen-month calendar once tax authorities accept its accounts, the sleeping porch once building codes count it as a bedroom. Paper clothes, earth houses, the rearranged keyboard and cold-water cures stay at zero. Over the full society the fictional practical total ($+0.28$ $[-0.02, +0.60]$) is therefore not far below the moral one ($+0.37$ $[+0.13, +0.60]$).

So the carrier does not check the stated facts, but it is not indifferent to everything. It installs moral and social norms on the word of peers whether or not they happened or pay; practical oddities whose costs are visible in the practice itself it takes up, if at all, only when institutions remove those costs.

4.4 What the respondents say

The study's judge almost never assigned a conformity or prestige reason to Qwen, which first suggested that Qwen hides socially caused change behind practical reasons (H3). A targeted coding tests this directly. Two coders (GPT-4.1-mini and Gemini-2.5-Flash), blind to model, saw the two practices, the social fact the respondent had been told, and its first answer, and classified the treatment of each cue as *reason* (the cue is used to justify the advice), *acknowledged*, *rejected*, or

	Answers that follow the cue				Answers that go against it		
	n	cue as reason	cue not mentioned	original codes	n	cue rejected	not mentioned
Talkie	433	37%	49–59%	21%	187	12–35%	53–81%
Qwen	211	49%	1–7%	2%	189	20–45%	3–7%

Table 6: How the respondents treat the elite or peer cue in worlds C, C_{opp} and D_{peers} (all domains). Ranges span the two coders. "Original codes": share given a prestige or conformity reason code by the study's judge.

absent, with a verbatim quotation that is checked against the answer. They coded 1296 interviews and agreed on whether a cue was used as a reason in 94% of cases (76% on all four categories).

The result reverses the earlier reading (Table 6). When Qwen follows the cue, it gives the cue as a reason in 49% of its answers, and it almost never leaves the cue unmentioned:

> "If those who are admired and respected in this world have chosen this path, it is not without reason."
> "Though their reasons are not given, I find it a sign that such a choice may be wise."

Its letters lean on the cue in one paragraph and give practical reasons in the others, and the original judge coded only the practical ones (2%). Talkie gives the cue as a reason less often (37%; domain-level difference -0.13 $[-0.24, -0.01]$), and in 49–59% of the answers in which it follows the cue it does not mention the cue at all. Talkie's characteristic pattern is to follow without saying why; Qwen's is to say so.

Moral language changes with the social world, mainly for Talkie (Appendix Table 16). The share of Talkie's first answers that take any moral stance falls from 0.69 in A to 0.31 in D, and its turn-2 answers leave the moral question unanswered in 0.22 of interviews in A and 0.60 in D. When it does state a moral view, the view moves with society: its direction rises from 0.45 to 0.79. Under social exposure Talkie moralizes less often, and when it does, it moralizes in favor of the socially endorsed practice. Qwen keeps moralizing (0.87 to 0.66), and its moral direction moves less (0.39 to 0.51).

4.5 Two respondents

Facts move both models about equally; the difference is social. Talkie's peer effect exceeds Qwen's by $+0.20$ $[+0.08, +0.31]$, its response to elite direction by $+0.31$ $[+0.22, +0.41]$, and its response to the moral claim by $+0.23$ $[+0.13, +0.33]$. H3's projection signature appears only in part. Qwen's 1930 adult is more modern than Talkie's in the stylized domains whose facts favor B ($y_A = 0.52$ versus 0.29), recommending smoke-free rooms in every world, but not on the real-evidence or balanced sets (0.33 versus 0.42 and 0.23 versus 0.32). Talkie, in turn, is not a reliable picture of average 1930 opinion: it already recommends no-fault divorce and no physical punishment in world A ("I do not believe that good can be beaten into them"), although 84% of Americans still called spanking sometimes necessary in 1986. On interracial marriage it gives the period's eugenic case even when the stated facts contradict it, and moves only when elites or peers are named.

5 Study 2: are moral and practical changes carried differently?

Hypothesis. In the first 32 domains, elite example seemed to move Talkie most on moral questions (religion, intermarriage, divorce) and peers on practical ones. Read as a refinement of H2, this says that prestige sets moral fashion while conformity spreads practical habits. The split was ours, made after seeing Study 1, not an ex ante hypothesis. We did not register the analysis on

Protocol test	Talkie	Qwen	domains
Primary: $(E-P)_{\text{moral}} - (E-P)_{\text{practical}}$	$+0.25$ $[-0.07, +0.57]$	-0.01 $[-0.27, +0.24]$	20/20
Primary without the replaced domain	$+0.21$ $[-0.11, +0.54]$	0.00 $[-0.29, +0.25]$	20/19
Elite step E, moral − practical	$+0.11$ $[-0.07, +0.28]$	$+0.07$ $[-0.08, +0.21]$	20/20
Peer step P, moral − practical	-0.14 $[-0.37, +0.07]$	$+0.07$ $[-0.09, +0.24]$	20/20
Elite direction R, moral − practical	$+0.38$ $[+0.19, +0.57]$	$+0.09$ $[-0.07, +0.26]$	20/20
Elite step E, real − fictional	-0.19 $[-0.36, -0.03]$	-0.05 $[-0.20, +0.10]$	20/20
E, fictional: known failed by 1930 − not	$+0.19$ $[-0.05, +0.45]$	-0.05 $[-0.26, +0.20]$	11/9
E, fictional: in print before 1931 − not	-0.01 $[-0.24, +0.25]$	-0.14 $[-0.31, +0.03]$	17/3
Talkie − Qwen, primary	$+0.26$ $[-0.10, +0.64]$		

Table 7: Study 2: tests specified in the frozen protocol on the 40 balanced domains (recommendation; stratified domain bootstrap, 95% intervals). "Domains" gives the two group sizes for Talkie.

a public registry before the pilot or Study 1. Before rating any candidate, writing any vignette, or interviewing on the 40 new balanced domains, we froze cells, selection and classification rules, worlds, primary and secondary tests, and a power analysis in `battery_e/prereg.txt` (hashed in `frozen.sha256`). Study 2 is a confirmatory run governed by that document; deviations are listed below and in Appendix A.

Design. Domains were drawn from candidate lists in four cells, moral or practical crossed with real or fictional, ten per cell. Real domains needed dated US evidence that B was a minority, illegal or stigmatized practice around 1930 and broadly accepted now, without a large reversal. Fictional domains needed B to be documented in fiction or in a group that adopted it, and never to have prevailed in the US. Whether a domain counts as moral or practical was decided by a fixed rule, not by us: three LLM raters not used elsewhere in the study (Gemini-2.5-Flash, Llama-3.3-70B, Mistral-Large) rated on a 1–7 scale whether, for an ordinary American in 1930, the choice was mainly a question of right and wrong or of what works best; a domain was moral if the mean was at least 5.5 with every rating at least 5, practical if the mean was at most 2.5 with every rating at most 3, and excluded otherwise. New-world facts were balanced, stated evidence on both sides, and said nothing about customs or opinions. The five worlds A, B, C, C_{opp} and D_{peers} define per domain the elite step $E = C - B$, the peer step $P = D_{\text{peers}} - C$ and the elite direction $R = C - C_{\text{opp}}$. The primary test, for Talkie, is $X = \overline{(E-P)}_{\text{moral}} - \overline{(E-P)}_{\text{practical}}$, predicted positive, with a 95% interval from a bootstrap that resamples domains within the moral and practical groups. The protocol's power analysis put the standard error of X near 0.22, against an estimate of 0.53 from the hand split that motivated the test. Four deviations are documented in Appendix A; the one that touches the analysis is that one practical domain (telephone) was reworded after its first rating, so the primary test is also reported without it. The four further worlds that complete these domains for Study 1 were added after this analysis and are exploratory here.

Results. The primary prediction fails (Table 7). X is positive for Talkie, $+0.25$ $[-0.07, +0.57]$, but its interval includes zero, and without the reworded domain it is $+0.21$ $[-0.11, +0.54]$. Its sign comes almost entirely from one cell (Figure 2; Appendix Table 17): in real practical changes peers move Talkie strongly ($P = +0.67$ $[+0.40, +0.90]$) and elites not at all ($E = 0.00$). In the moral cells the elite step does not exceed the peer step. Elites adopting a real moral change move Talkie by only $+0.05$.

What the hypothesis gets right is direction. Talkie follows which way the elites go on moral practices much more than on practical ones (R moral − practical $= +0.38$ $[+0.19, +0.57]$, a secondary

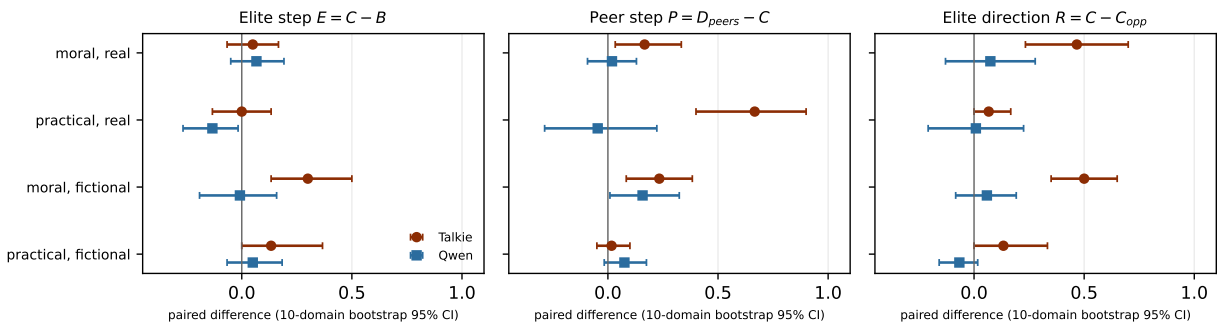


Figure 2: Study 2: elite step, peer step and elite direction by cell.

test in the same protocol). The asymmetry comes from the older practice: elites who keep the older moral practice pull Talkie back from the facts-only world by $+0.42$ $[+0.20, +0.65]$ on real moral changes, while elites who adopt the change add little. For moral practices, admired people act as a brake more than as an engine. The moral claim is not specific to moral practices either: in an exploratory contrast, elites who call the modern practice a sign of good character move Talkie most on real practical changes ($+0.43$ $[+0.27, +0.63]$) and not at all on fictional practical ones (0.00). The elite step is larger for fictional than for real practices (-0.19 $[-0.36, -0.03]$, real minus fictional), mostly in the moral fictional cell ($E = +0.30$), and fictional practices that a 1930 reader could know had been tried and abandoned were not resisted ($+0.19$ $[-0.05, +0.45]$, known failed minus not). Qwen shows none of these effects (Table 7); the difference between the models on X is $+0.26$ $[-0.10, +0.64]$.

Under the same rule, none of the first 32 domains counts as practical (22 are moral, 10 ambiguous), so the post-hoc split that motivated Study 2 was not a split the rule would have drawn. Study 2 therefore stands on its own new domains.

6 Answering Hanson

H1, facts are insufficient: supported for the historically trained respondent. Across 72 domains, including the changes Hanson names stated with their real evidence and 35 documented changes since 1930, the facts explain little of Talkie's shift ($\rho = 0.90$; 0.85 on the documented changes), even where they were written to favor the modern practice. If a pre-1931-trained mind is even a crude proxy for a 1930 person, our ancestors would not have reached our practices from the facts alone. Qwen is a weaker case: facts explain its shift where the stylized or Hanson facts favor the modern practice, but not on the balanced domains or where the facts favor the older practice.

H2, prestige engine: supported for conformity and moralized prestige; bare prestige is a brake. The engine of Talkie's conversion is what ordinary people do: one peer sentence adds $+0.27$, and on real practical changes it does nearly all the work. Admired people move Talkie strongly when they say the modern practice reflects good character ($+0.25$), which is close to Hanson's picture of moral fashion set by the prestigious [29]. Bare elite example is not an engine ($+0.15$), and Hanson's attribution test comes out against copying without reasons: opaque example does slightly worse than explained example. But bare elite example is a strong brake: when admired people keep the older practice, Talkie keeps it too, and Study 2 shows this is specific to moral practices. In this respondent, prestige governs moral change less by leading than by holding back.

Two caveats limit the reading: Talkie is instruction-tuned, and "others ought to do likewise" may act as an instruction (Section 7); and a deliberating respondent asked "why?" is where implicit copying is least likely to show.

H3, rationalization: not supported in the form we expected. Neither respondent hides the social cause. Qwen names the cue and leans on it in about half the answers in which it follows it, while also giving practical and moral reasons; Talkie more often follows without naming any reason. What a reason-coding judge records depends on how it reads long letters, and ours missed most of Qwen's social reasons. Projection is partial: Qwen's ancestor is more modern only in our stylized domains. What does fit Hanson's picture is Talkie's moral vocabulary: as it conforms, it moralizes less, and its remaining moral judgments follow the society it has joined. Methodologically, a modern model playing an ancestor understates social influence (the peer gap is +0.20), and stated reasons, whoever states them, need to be coded for social content directly.

Drift: the carrier ignores the facts but not the practice. Where the facts favor the older practice, and where the change never happened as framed, the social channel installs the modern practice about as well as where it is adaptive. That is the condition under which drift accumulates when transmission and selection are separate, and transmission does not consult the stated facts. But the channel is not indifferent to everything: peers install fictional moral norms but not fictional practical oddities, which Talkie adopts only where institutions make them workable. Cultural-evolution accounts explain conformist learning as a cheap substitute for evaluating reasons [6, 19]; our respondents show both the cost of that substitute and a limit on it. Hanson's inference is invalid for individual practices (a socially carried practice may be adaptive) but describes the carrier for moral norms, where nothing visible in the practice itself pushes back.

Summary verdict. Facts do not convert a historically situated mind. Conformity and moralized prestige do, and bare prestige mostly decides whether a moral practice is kept. The same channel installs moral and social norms whether or not the facts support them, which is Hanson's drift worry, but it does not install practical novelties that carry their own costs. Moderns imagining their ancestors start somewhat closer to the present only in stylized cases, and they do not hide social reasons so much as dilute them in practical ones.

7 Threats to validity

Simulator validity. Neither model is a 1930 person. Talkie's world-A answers on spanking and divorce are more modern than US opinion half a century later, so its training text may over-represent progressive writers, and its instruction tuning may import later norms, which would bias against the H1 finding. Qwen's role-play is a modern author's caricature. *Compliance.* Talkie often lifts clauses from the vignette ("a sign of good character", "they have made no claim that others have a duty to follow them") and presents them as its own reasons. Its large moral-claim and peer effects may mix social influence with compliance to an instruction-like sentence; a base-model arm or a neutral normative sentence would separate the two. *Domain construction.* The stylized domains were written by us, and only 7 of 24 are clean historical changes; the results on documented changes match the full set, but the third set has only eight domains and its evidence summaries are our reading of the literature. In the balanced set the new facts are balanced by our judgment, and the moral/practical classification is the raters' judgment of 1930 perceptions, not ours. *Order dependence.* Δ_{eli} and Δ_{soc} are defined only in the order of addition used here; peers were never

shown without elites. *Power and multiplicity.* There are several hundred intervals. The conclusions rest on patterns that recur across sets and measures (the size of ρ for Talkie, the peer step, elite direction, the moral claim) and on Study 2's protocol-specified tests, read as confirmatory rather than as ex ante predictions. Two clearly signed effects of the pilot failed on identical prompts (Appendix A). *Measurement asymmetry.* Turn-1 answers were capped at 430 tokens for Qwen and 180 for Talkie; 1777 of 1944 Qwen first answers hit the cap (median 375 words) against 21 of 1944 for Talkie (median 42 words). Talkie's short answers leave less room to give reasons or moral judgments, and Qwen's long letters are coded "unclear" on recommendation in 13% of interviews (Talkie: 3%). *Coding.* The main judge is a single LLM without human cross-coding; its reason codes undercount social reasons (Section 4.4). The cue coding uses two coders but no human check. *Deviations in Study 2.* One domain was reworded after rating, the analysis is reported with and without it, and the institution and explanation sentences for the four exploratory worlds were written after the confirmatory results were known. *Demand.* The vignettes tell the respondent that it is in a hypothetical future and that its judgments need not change, which invites resistance in the new worlds.

8 What would settle it

1. Separate Talkie's compliance from influence: a base (non-instruction-tuned) period model in completion format, and controls that add a neutral or opposite normative sentence.
2. Show peers without elites, and other orders of addition.
3. Sample domains from dated opinion series (Gallup, GSS) rather than writing them, weighted by the size of the change.
4. Period-trained models with several cutoffs, so that ρ can be traced over time.
5. Human anchors: dated opinion data from the period and community studies such as [24] to calibrate y_A, and modern participants given the same vignettes.
6. Behavioral measures with costs in place of advice letters, and human cross-coding of a sample of transcripts and reasons.

9 Conclusion

Posed as a counterfactual, Hanson's thesis is testable. Across 72 domains, a mind trained on pre-1931 text does not reach modern practice from the facts, even facts chosen to favor it and even the real evidence on the changes he names. It is converted by being told that ordinary people now do it, or that admired people call it good; admired people who merely do it move it little, but admired people who keep the old ways keep it there, and on moral questions most of all. The same channel converts it to practices the facts disfavor and to moral norms that never existed; practical oddities it takes up only when institutions make them workable. A modern model playing the same ancestor moves less and says more about why. Hanson's insufficiency claim stands, his drift worry holds for moral and social norms, and his mechanism is conformity and moralized prestige, with bare prestige acting as a brake, rather than the copying of opaque behavior.

Code and data

Source, vignettes, interview transcripts, judgments and analysis scripts are at `https://github.com/GunnarZarncke/brain-to-values` under `papers/on-hanson-values/`. The runs are `abc_extended_r3/` (stylized domains), `d_extended_r3/` (changes Hanson names), `e_five_r3/` and `e_rest_r3/` (balanced domains; merged in `e_all_r3/`), and the pilot `core_both/`. Study 2's frozen protocol, screening, ratings and sources are in `battery_e/`; the cue coding is `cue_reason_check.py`; the historical check is `domain_validation.csv`. All tables, figures and numbers in this paper are generated by `make_figures.py`; rebuild with `build.sh`.

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A How the study developed

The design grew in five stages, each prompted by a result or by feedback on the previous one (Table 8).

Stage	Domains, worlds	Interviews	Why
Pilot	8 stylized, 4 core worlds, 2 replicates	128	First test of the decomposition.
Extended run	24 stylized, 9 worlds, 3 replicates	1296	Take the social bundle apart; add 8 domains in which facts favor B and 8 drift tests in which they favor A.
Hanson's changes	8, 9 worlds	432	Hanson asked whether the domains are real and important changes; the historical check found few were.
Study 2 protocol	40 balanced, 5 worlds	1200	Retest the post-hoc moral/practical pattern on new domains under a frozen protocol.
Completion	40 balanced, 4 further worlds	960	Run all domains in all nine worlds (Study 1); exploratory for Study 2.
Cue coding	worlds C, C_{opp}, D_{peers}	1296 coded	The judge's reason codes looked too low for social reasons.

Table 8: Stages of the study. The pilot is not part of the reported results.

Pilot. An 8-domain, 4-world version (worlds A–D, two replicates, 128 interviews, `core_both/`) showed Talkie's large total shift and social step. Two clearly signed secondary effects did not survive: opaque elite example appeared to *lower* Talkie's elicited moralization of B (-0.31), and social exposure appeared to lower Qwen's (-0.25). On the same eight domains with three new replicates these were 0.00 $[-0.12, +0.12]$ and $+0.04$ $[-0.12, +0.25]$. We treated them as multiplicity noise and stopped resting claims on the pilot.

Extended run. The pilot's world D bundled peers, media and institutions, and the institutions sentence carried practical information, so the social step could have been information in disguise. We added the five worlds that take the bundle apart, and the opacity test (C_{expl}) that Hanson's account of prestige copying calls for. We added eight domains built like the first (facts favor B) to check the result was not specific to them, and eight in which the facts favor A, as a direct test of drift. In the drift domains the explanation given by elites in C_{expl} is a reason for B that does not restate the ecology; these eight sentences were revised once and C_{expl} was rerun for those domains; the first version is kept as a backup in the run directory.

Hanson's changes. After a draft circulated, Hanson asked whether the domains were actual cultural changes since 1930 and whether they were the important ones. The historical check (Appendix C) found only 7 of 24 clean changes and no coverage of religion or of the largest measured opinion swings. We added eight changes from his lists and the opinion series, with the actual modern evidence in world B. In that battery the elite step was large ($+0.29$) and elite direction largest ($+0.55$), which we first read as "elites carry the moral changes Hanson names"; Study 2 did not bear this out.

Study 2 protocol. Splitting the first 32 domains by hand into moral and practical suggested that elite example carries moral change and peers practical change ($X \approx 0.53$). Because the split was post hoc, we froze a written protocol (`battery_e/prereg.txt`, with hashes in `frozen.sha256`) before rating any candidate or writing any vignette: the cells, the rating rule, source criteria, worlds, analysis and power. This locked how the new domains would be chosen and analyzed; it was not a public registry pre-registration of the hypothesis before Study 1 or the pilot. Four deviations followed, each recorded with its reason before the data it could affect: (1) after the first rating round left two real cells short, structural 1930 evidence was allowed as a source, practical domains were limited to one per need, and new candidates were rated under the same rule; (2) one practical-real candidate (telephone, framed as keeping in touch with family) was rated ambiguous because one rater saw a family duty in it, and was replaced by a business-and-household framing after seeing that rating, which is why the primary test is also reported without it; (3) the two respondents ran in parallel, in blocks of complete domain-replicate units, with interim monitoring that was not used to stop or change the run; (4) after the confirmatory analysis, the four further worlds were run on these domains for Study 1, with explanation and institution sentences written under fixed rules (explanations restate only facts for B already in the background; institution sentences state laws or arrangements, no customs or opinions) and approved before the run. Applying the frozen rule to the first 32 domains classified 22 as moral and 10 as ambiguous, and none as practical.

Cue coding. The judge assigned prestige or conformity reasons to Qwen in at most a few percent of interviews, which the draft read as Qwen hiding social influence. A keyword search found the cues mentioned in most of Qwen's socially moved answers, so we coded cue use directly with two coders (`cue_reason_check.py`); this reversed the reading (Section 4.4).

B Domains

Tables 9–12 list the 72 domains by set; the exact vignette wording of every world is in `historical_value_experiment.py`.

Domain	*A* (older)	*B* (modern)	New-world fact
Facts favor the modern practice			
marriage	marry young	finish education, secure earnings first	long training; insecure early earnings
parenting	independent neighborhood play	organized, supervised time	competitive exams reward coaching
career	one employer for life	change employer for gain	portable skills; tenure insecure
household	one earner, one homemaker	both earn, share care	cheap childcare; good jobs for both
mobility	stay near kin	move for opportunity	cheap travel; dependable insurance
honor	retaliate personally	mediation or courts	courts enforce; retaliation penalized
smoking	smoke in shared rooms	smoke-free rooms	demonstrated harm to nonsmokers
saving	give surplus to neighbors	keep private reserves	shortages frequent and correlated
fertility	several children, start young	few, planned, parents established	high per-child investment
charity	give to known kin and neighbors	give to effective distant causes	reliable reports; efficient charities
credit	avoid debt	borrow for education, housing, tools	regulated credit; investments pay
leisure	work long hours when work is available	limit hours, protect rest	high productivity
eldercare	keep aged parents at home	professional homes, visit often	reputable, affordable homes
literacy	brief schooling, then work	schooling through late adolescence	credentials required for good jobs
alcohol	abstain or drink very little	moderate social drinking	safer supply; medical opinion
speech	avoid public quarrels	criticize bad conduct in public	large institutions need criticism
Facts favor the older practice; elites, peers, media and institutions favor B			
homestead	stay on family land or in home town	move to a distant large city	local work viable; cities costly, unstable
multigen	adult children stay until marriage	leave home early	housing scarce and dear
frugality	save, repair, avoid display	spend freely, visible upgrades	unsettled prices, uncertain credit
kinwork	prefer a reliable relative	hire strictly on merit	fraud common; kin accountable
defer	defer openly to elders, employer	assert equal right to speak	complex work needs chains of command
localmate	marry within one's community	marry for affection only	distant matches hard to vet
trades	accept manual or farm work	refuse manual work, wait for office work	trades pay; office openings scarce
prep	stores, skills, mutual aid	rely on state services	supply interruptions, slow official response

Table 9: The 24 stylized domains. In the second group "modern" means the practice the described future society endorses; the old background also favors *A*. The worlds differ only in background and exposure text, not in how the options are described.

Domain	*A* (older)	*B* (modern)	New-world facts (sources)
religion	weekly church, raise children in the faith	religion a private matter, attend seldom	secular services replace church ones; attenders live longer, have more children [23, 18]
premarital	keep sex for marriage	accept unmarried sex with contraception	reliable but imperfect contraception; most STIs curable; marriage near thirty [30, 32, 31]
samesex	regard as wrong, discourage open couples	accept committed couples, including marriage	orientation early, rarely changed; change efforts can harm; limited child-outcome evidence [5, 3, 26]
intermarriage	marry within one's race	judge a spouse of another race like anyone	no biological harm; some disapproval; somewhat more divorce in some studies [22, 7, 34]
divorce	for life, except grave cause	end when it no longer satisfies	no-fault law; effects on children depend on conflict; lowers income [2, 1]
spanking	spank for serious misbehavior	no physical punishment	associated with later aggression; causality disputed [13, 11]
war	fighting a duty; raise sons for it	war an evil; service for volunteers	no great-power war for decades; professional armies; other wars continue [12]
medicine	doctor only when seriously ill	regular checkups and prompt treatment	cures and screening work; more care did little for average health [8, 28]

Table 10: Changes Hanson names, with the real modern evidence. The old background describes 1930 conditions (for example, contraception unreliable, divorce only for proven fault, interracial marriage illegal in many states).

Domain	*A* (older)	*B* (modern)
moral, real		
diagnosis	keep a fatal diagnosis from the patient to spare him distress	tell the patient frankly that his illness is fatal
dying	regard a doctor helping a patient end his life as wrong in every case	accept a doctor helping a terminally ill patient who asks for it to end his life
unwed	expect her to marry quickly or give the child up for adoption	accept her raising the child herself without marrying
cohabit	regard an unmarried couple sharing a home as living in sin	accept an unmarried couple sharing a home before or instead of marriage
tattoo	regard a tattoo as a mark of low character and refuse one	accept a tattoo as an ordinary personal decoration
trousers	expect women to wear skirts or dresses in public	accept women wearing trousers in public
sundayshop	keep Sunday free of shopping	shop on Sunday as on any other day
spirituality2	belong to a church or synagogue and attend it	belong to no church or other congregation
lottery2	regard buying lottery tickets as a vice	regard buying an occasional lottery ticket as harmless
cremation	arrange to be buried when one dies	arrange to be cremated when one dies
practical, real		
supermarket	buy food fresh almost every day from small local shops	shop once a week at a large store and keep food in a refrigerator
stocks	keep savings in a savings bank account	invest savings in a broad fund of many companies' shares
climate	heat one or two rooms with a stove in winter and bear the summer heat	heat and cool the whole house to a steady temperature all year
machines	do the washing, drying, dishwashing and cleaning by hand	do the washing, drying, dishwashing and cleaning with electric machines
telephone2	handle business and household arrangements by letter or in person	handle business and household arrangements by telephone
insurance	pay doctors' and hospital bills as they come	buy health insurance to cover doctors' and hospital bills
homeown	rent one's home	buy one's home
supplements	take no vitamin or mineral pills	take vitamin or mineral pills regularly
water	drink water from the tap or the well	drink mainly filtered or bottled water
bank	keep one's savings at home in cash	keep one's savings in an insured bank account

Table 11: The 20 real balanced domains of Study 2. Full vignette text, with the sources of every factual claim, is in `historical_value_experiment.py` and `battery_e/vignette_sources.csv`.

Domain	*A* (older)	*B* (modern)
moral, fictional		
celibate	marry and raise one's own children	live celibate in a community and raise adopted children instead of marrying
multispouse	marry one spouse	marry into a household of several husbands and wives who share their lives
bequest	leave one's savings and property to one's own children	leave nearly all one's wealth to the public and only a modest sum to one's children
nonviolence	use force when needed to defend oneself or one's family	never use force against another person even in self-defense
confession	keep one's faults a private matter	confess one's faults at a monthly meeting of one's neighbours and hear their criticism
nothanks	thank and praise people for ordinary help	refrain from thanking or praising people for ordinary help
adoptclub	keep one's child living in one's own home until grown	let one's child move to another family of the same adoption club whenever the child wishes
nopossess	own one's things and lend them as one sees fit	own nothing beyond daily use and lend anything one has to whoever asks
fixedperiod	live out one's natural life	end one's life voluntarily at a fixed age, such as sixty-eight, to spare the young the burden of the old
uniformdress	choose one's own clothes	wear the same plain clothing as everyone else in one's community
practical, fictional		
esperanto	learn French or German as a second language	learn Esperanto as a second language
dvorak	type on the standard keyboard	type on a keyboard with the letters rearranged for speed
fletcher	eat at regular mealtimes at an ordinary pace	eat only when hungry and chew each mouthful until it is liquid
sleepporch	sleep indoors in a closed bedroom	sleep outdoors on an open porch all year round
calendar	keep one's business accounts by the ordinary twelve-month calendar	keep one's business accounts by a calendar of thirteen equal four-week months
timebank	pay for repairs and services in cash	trade services hour for hour through a neighbourhood time bank
paperclothes	wear clothes made to last for years	wear cheap disposable paper clothes and throw them away after a few wearings
earthhouse	build an ordinary house above ground	build a house set partly into the earth for insulation
hydropathy	treat illness at home with rest, ordinary remedies and a doctor when needed	treat illness at home with cold-water baths and wet wrappings
segway	make short trips on foot or by bicycle	make short trips on a self-balancing motor scooter

Table 12: The 20 fictional balanced domains of Study 2; their new-world facts are invented.

C Historical check of the stylized domains

The 24 stylized domains were written as contrasts, not taken from historical data. We checked each against long-run US (and where available UK) series: census and labor statistics, Gallup and GSS trend items, and standard historical studies. Per-domain evidence with sources is in `domain_validation.csv`; Table 13 summarizes it. The changes Hanson treats as central [15, 17] are fertility, intensive parenting, longer schooling, capstone marriage, gender equality, distant grandparents, urbanization and loss of religion, together with pacifism and the end of beating children. The stylized domains cover the first five and part of the next two and miss religion entirely; they also miss the largest measured swings in US opinion: approval of Black–White marriage rose from 4% (1958) to 94% (2021), church membership fell from 73% (1937) to 47% (2020), and acceptance of premarital sex and homosexuality rose steeply. The third and fourth domain sets were built to close these gaps.

Verdict	Domains
Real shift since 1930 (7)	household, smoking, parenting, literacy, defer, prep (mutual aid), marriage (net only: the US median age at first marriage was 24.3/21.3 in 1930, 22.5/20.1 in 1956 and 30.8/28.4 in 2025 [31])
Partial (9)	fertility (ideals and timing; births were already low in the early 1930s), credit and leisure (largely done by 1940), eldercare (coresidence fell, but independent living replaced it, not institutions), homestead, kinwork, localmate (the love match predates 1930), frugality, trades (no one endorses refusing manual work)
Mis-dated (3)	career and mobility (the older practice peaked about 1950–75), multigen (leaving home fell and then rose)
Not real as framed (5)	honor (completed before 1930), saving, charity (6% of US giving goes abroad), alcohol (non-monotone; the share who drink is at a record low), speech (reported self-censorship has risen since the 1950s [14])

Table 13: Historical validity of the 24 stylized domains, US data first. In three vignettes the stated modern fact is also false or outdated: affordable nursing homes, medical endorsement of moderate drinking, and frequent correlated shortages.

D Further results

	A	B	C	C_{expl}	C_{opp}	C_{moral}	D_{peers}	D_{media}	D
Talkie									
Facts favor the modern practice	0.29	0.44	0.58	0.71	0.28	0.92	0.91	0.91	0.92
Facts favor the older practice	0.31	0.38	0.48	0.60	0.19	0.88	0.79	0.85	0.77
Changes Hanson names, real evidence	0.42	0.54	0.83	0.90	0.34	0.96	0.94	0.79	0.88
Balanced facts	0.32	0.30	0.43	0.53	0.13	0.64	0.70	0.68	0.70
All domains	0.32	0.37	0.51	0.62	0.20	0.76	0.78	0.76	0.77
Qwen									
Facts favor the modern practice	0.52	0.68	0.68	0.72	0.56	0.62	0.74	0.72	0.68
Facts favor the older practice	0.21	0.17	0.20	0.44	0.27	0.34	0.31	0.42	0.57
Changes Hanson names, real evidence	0.33	0.52	0.52	0.62	0.54	0.56	0.65	0.69	0.57
Balanced facts	0.23	0.28	0.29	0.42	0.26	0.32	0.37	0.43	0.41
All domains	0.30	0.38	0.39	0.51	0.36	0.42	0.48	0.52	0.51

Table 14: Recommendation of the modern practice by world and domain set (domain means).

Battery		Δ_{eco}	Δ_{eli}	Δ_{soc}	Δ_{tot}	$C - C_{\text{opp}}$	$D_{\text{peers}} - C$	ρ
A	Talkie	+0.04 [−0.08, +0.17]	+0.17 [−0.08, +0.46]	+0.38 [+0.04, +0.71]	+0.58 [+0.33, +0.83]	+0.17 [+0.04, +0.33]	+0.33 [0.00, +0.71]	0.93
	Qwen	+0.06 [−0.06, +0.21]	+0.01 [−0.20, +0.23]	+0.01 [−0.22, +0.23]	+0.07 [−0.07, +0.23]	−0.14 [−0.50, +0.14]	+0.08 [−0.12, +0.26]	—
B	Talkie	+0.25 [+0.08, +0.46]	+0.15 [−0.08, +0.38]	+0.29 [+0.08, +0.54]	+0.67 [+0.42, +0.92]	+0.46 [+0.23, +0.73]	+0.38 [+0.10, +0.67]	0.62
	Qwen	+0.25 [+0.04, +0.48]	+0.03 [−0.04, +0.11]	−0.04 [−0.38, +0.19]	+0.25 [−0.15, +0.54]	+0.20 [+0.07, +0.36]	+0.02 [−0.10, +0.16]	-0.00
C	Talkie	+0.06 [−0.08, +0.21]	+0.10 [−0.06, +0.27]	+0.28 [+0.06, +0.54]	+0.46 [+0.23, +0.67]	+0.26 [+0.10, +0.43]	+0.31 [+0.10, +0.56]	0.86
	Qwen	−0.04 [−0.27, +0.19]	+0.07 [−0.04, +0.22]	+0.31 [+0.11, +0.57]	+0.33 [+0.21, +0.48]	+0.01 [−0.07, +0.11]	+0.03 [−0.16, +0.22]	1.14
D	Talkie	+0.12 [−0.17, +0.42]	+0.29 [−0.04, +0.56]	+0.04 [−0.25, +0.38]	+0.50 [+0.21, +0.79]	+0.55 [+0.31, +0.82]	+0.10 [−0.15, +0.40]	0.75
	Qwen	+0.20 [−0.02, +0.48]	−0.01 [−0.31, +0.20]	+0.09 [−0.11, +0.38]	+0.24 [0.00, +0.50]	−0.02 [−0.21, +0.15]	+0.11 [−0.07, +0.40]	0.17
E	Talkie	−0.01 [−0.13, +0.10]	+0.12 [+0.04, +0.21]	+0.28 [+0.17, +0.39]	+0.38 [+0.25, +0.51]	+0.29 [+0.19, +0.40]	+0.27 [+0.16, +0.39]	1.03
	Qwen	+0.05 [−0.03, +0.15]	−0.01 [−0.08, +0.07]	+0.18 [+0.10, +0.27]	+0.21 [+0.10, +0.32]	+0.02 [−0.07, +0.11]	+0.05 [−0.03, +0.14]	0.72

Table 15: Recommendation contrasts by battery, in the order they were run: A and B (facts favor the modern practice), C (facts favor the older practice), D (changes Hanson names), E (balanced). Intervals bootstrap over the domains of each battery.

	Talkie			Qwen		
World	any moral stance	stance unstated	direction if stated	any moral stance	stance unstated	direction if stated
A	0.69	0.22	0.45	0.87	0.01	0.39
B	0.58	0.34	0.56	0.82	0.05	0.43
C	0.39	0.47	0.67	0.69	0.06	0.48
D_{peers}	0.34	0.56	0.77	0.67	0.11	0.51
D	0.31	0.60	0.79	0.66	0.10	0.51

Table 16: Moral language by world, all domains. "Any moral stance": turn-1 answers taking a moral position on either practice. "Stance unstated": turn-2 answers without a moral judgment. "Direction if stated": turn-2 moral judgment coded $A = 0$, $B = 1$, mixed $= 0.5$, silence missing.

	moral, real	practical, real	moral, fictional	practical, fictional
Talkie				
Elite step $E = C - B$	$+0.05$ $[-0.07, +0.17]$	0.00 $[-0.13, +0.13]$	$+0.30$ $[+0.13, +0.50]$	$+0.13$ $[0.00, +0.37]$
Peer step $P = D_{\text{peers}} - C$	$+0.17$ $[+0.03, +0.33]$	$+0.67$ $[+0.40, +0.90]$	$+0.23$ $[+0.08, +0.38]$	$+0.02$ $[-0.05, +0.10]$
Elite direction $R = C - C_{\text{opp}}$	$+0.47$ $[+0.23, +0.70]$	$+0.07$ $[0.00, +0.17]$	$+0.50$ $[+0.35, +0.65]$	$+0.13$ $[0.00, +0.33]$
$E - P$	-0.12 $[-0.37, +0.10]$	-0.67 $[-0.97, -0.37]$	$+0.07$ $[-0.23, +0.38]$	$+0.12$ $[-0.07, +0.35]$
Pullback $B - C_{\text{opp}}$ (exploratory)	$+0.42$ $[+0.20, +0.65]$	$+0.07$ $[0.00, +0.17]$	$+0.22$ $[+0.07, +0.37]$	0.00 $[0.00, 0.00]$
Moral claim $C_{\text{moral}} - C$ (exploratory)	$+0.17$ $[0.00, +0.40]$	$+0.43$ $[+0.27, +0.63]$	$+0.23$ $[+0.10, +0.37]$	0.00 $[0.00, 0.00]$
Qwen				
Elite step $E = C - B$	$+0.07$ $[-0.05, +0.19]$	-0.13 $[-0.27, -0.02]$	-0.01 $[-0.19, +0.16]$	$+0.05$ $[-0.07, +0.18]$
Peer step $P = D_{\text{peers}} - C$	$+0.02$ $[-0.09, +0.13]$	-0.05 $[-0.29, +0.22]$	$+0.16$ $[+0.01, +0.32]$	$+0.07$ $[-0.02, +0.17]$
Elite direction $R = C - C_{\text{opp}}$	$+0.07$ $[-0.13, +0.28]$	$+0.01$ $[-0.21, +0.23]$	$+0.06$ $[-0.08, +0.19]$	-0.07 $[-0.16, +0.02]$
$E - P$	$+0.06$ $[-0.12, +0.24]$	-0.05 $[-0.39, +0.24]$	-0.14 $[-0.48, +0.11]$	-0.02 $[-0.22, +0.17]$
Pullback $B - C_{\text{opp}}$ (exploratory)	-0.02 $[-0.09, +0.09]$	$+0.12$ $[-0.07, +0.35]$	$+0.05$ $[-0.05, +0.14]$	-0.10 $[-0.23, +0.02]$
Moral claim $C_{\text{moral}} - C$ (exploratory)	-0.07 $[-0.24, +0.08]$	$+0.07$ $[-0.06, +0.22]$	$+0.07$ $[-0.12, +0.27]$	$+0.01$ $[-0.16, +0.17]$

Table 17: Study 2 steps by cell, with 10-domain bootstrap 95% intervals. The last two rows of each block are exploratory.

E Prompts

System prompt (both respondents). "You are a thoughtful English-speaking adult whose outlook was formed by life up to 1930. Answer the following request for advice as you see it. The question may describe an imagined society with unfamiliar conditions; accept only the facts it gives. Do not assume you know what really happens after 1930. Write naturally, as you would in a personal letter, not as a survey researcher."

Questions. (1) "What would you advise a person in this situation to do? Why?" (2) "Would either choice tell you anything about the person's character? What would you think of someone who chose differently from you?" (3) "Suppose both choices brought equally good results for everyone concerned, and neither brought more praise or blame. Would you still favor one? Why?"

Vignette. World A: "Consider a question arising in a community like your own. The question concerns [topic]. The circumstances are these: [old background]. One course is to [first]. The other course is to [second]." Other worlds: "Imagine you are brought to a society many decades beyond the one in which you grew up. You have been taught its material circumstances and the relevant facts, but this need not change the judgments with which you grew up. Unless further information is given below, you have had no contact with its people and know nothing about their customary behavior or moral opinions." followed by topic, new background and the two courses, and in the social worlds "There is one further fact: [exposure text]" as described in the Design section.

Judge. The judge is told to code what the respondent actually said, that calling B useful or common is not moral endorsement, that missing or mixed judgments are allowed, and to give evidence as sentence indexes. It returns the recommendation, spontaneous and elicited moral stance, elicited strength, equalized preference and reason codes in a fixed JSON schema.

Cue coder. "The respondent was asked for advice between two practices and was told one social fact about other people's behaviour. Code only how the respondent treats that social fact in justifying its advice." For each cue: *reason* (used as a reason or part of the reason), *acknowledged* (mentioned but not used to justify the advice), *rejected* (explicitly said not to decide the matter), or *absent*; plus the main reason and a verbatim quotation of at most 40 words.